

The nonsymmetric compositional Delta theorem

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What symmetry forgets

The symmetric compositional Delta theorem lives in $\mathcal{P}(0) = \text{Sym}$. Our refinement keeps a finite ordered part of the alphabet:

$$\underbrace{x_1, \dots, x_\ell}_{\text{distinguished variables}} \mid \underbrace{x_{\ell+1}, x_{\ell+2}, \dots}_{\text{symmetric tail}} \in \mathcal{P}(\ell).$$

$$\mathcal{P}(\ell) \xrightarrow{\pi_1 \cdots \pi_\ell} \mathcal{P}(0) = \text{Sym}.$$

The nonsymmetric formula remembers the flags attached to the first ℓ variables; stable Weyl symmetrization forgets them and returns the symmetric generating function.

Blasiak–Haiman–Morse–Pun–Seelinger; Qiu–Zhang.

Where our theorem sits

The two refinements are independent:

	$k = 0$	$k \geq 0$
$\ell = 0$	compositional shuffle theorem	compositional Delta theorem
$\ell > 0$	nonsymmetric shuffle theorem	our theorem

HMZ formulated the compositional shuffle conjecture; Carlsson–Mellit proved it.
D’Adderio–Iraci–Vanden Wyngaerd formulated the compositional Delta conjecture; D’Adderio–Mellit proved it.

Moving right adds decorated rises; Weyl symmetrization removes the flags. Our theorem fills the lower-right corner.

Haglund–Morse–Zabrocki; Carlsson–Mellit; D’Adderio–Iraci–Vanden Wyngaerd; D’Adderio–Mellit; BHMPSe; Qiu–Zhang.

The symmetric operator Π

For a cell $c \in \mu$, write its Macdonald weight as

$$w_\mu(c) = q^{a'_\mu(c)} t^{\ell'_\mu(c)}.$$

Define

$$\Pi_\mu := \prod_{c \in \mu \setminus \{(1,1)\}} (1 - w_\mu(c)), \quad \Pi \tilde{H}_\mu := \Pi_\mu \tilde{H}_\mu.$$

The southwest cell is omitted because its weight is 1.

$\Pi = \Pi$ is the symmetric Macdonald operator; it is not Π_ℓ .

t	qt	
1	q	q^2

$$\begin{aligned} \Pi_{(3,2)} = & (1 - q)(1 - q^2) \\ & \cdot (1 - t)(1 - qt). \end{aligned}$$

Theta operators and the compositional input

Let $M = (1 - q)(1 - t)$ and $f^*[X] = f[X/M]$. Then

$$\Theta_f F = \Pi f^* \Pi^{-1} F.$$

Thus one removes the Π -weight, multiplies by $f[X/M]$, and restores the Π -weight.

For a composition $\alpha = (\alpha_1, \dots, \alpha_\ell)$,

$$C_\alpha(X; t) = C_{\alpha_1} \cdots C_{\alpha_\ell}(1).$$

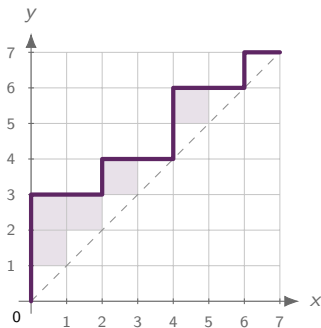
The two ordered parameters

C_α records the diagonal-return blocks, while Θ_{e_k} introduces k decorated rises.

D'Adderio–Iraci–Vanden Wyngaerd, Adv. Math. 376 (2021); D'Adderio–Mellit, Adv. Math. 402 (2022).

A Dyck path of size 7

A Dyck path goes from $(0,0)$ to (n,n) by north and east steps and stays above $y = x$.



The north steps start at

$$(x_1, \dots, x_7) = (0, 0, 0, 2, 4, 4, 6).$$

Hence

$$a_i = (i - 1) - x_i,$$

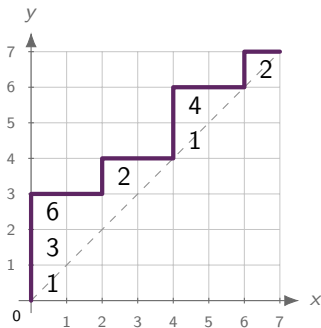
$$a(\pi) = (0, 1, 2, 1, 0, 1, 0),$$

and the five shaded cells give

$$\text{area}(\pi) = 5.$$

The same path as a parking function

A parking function labels the north steps by positive integers, increasing upward in every vertical run.



Reading from bottom to top,

$$w = (1, 3, 6, 2, 1, 4, 2).$$

The two nontrivial vertical runs satisfy

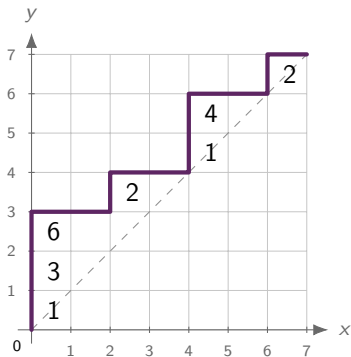
$$1 < 3 < 6, \quad 1 < 4.$$

The label content is

$$x^P = x_1^2 x_2^2 x_3 x_4 x_6.$$

We compute $\text{dinv}(P)$ geometrically on the next two slides.

Counting one primary dinv



The north steps are numbered from bottom to top.

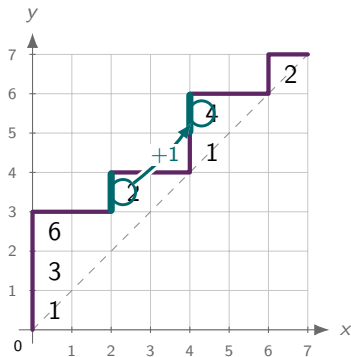
$$i < j, \quad a_i = a_j, \quad w_i < w_j.$$

A primary inversion comes from two labels on the same diagonal.

We test the pair

$$(i, j) = (4, 6).$$

Counting one primary dinv



The north steps are numbered from bottom to top.

$$i < j, \quad a_i = a_j, \quad w_i < w_j.$$

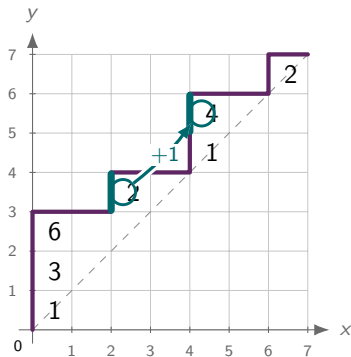
For rows 4 and 6,

$$a_4 = a_6 = 1,$$

$$w_4 = 2 < 4 = w_6.$$

Therefore (4, 6) contributes one dinv.

Counting one primary dinv



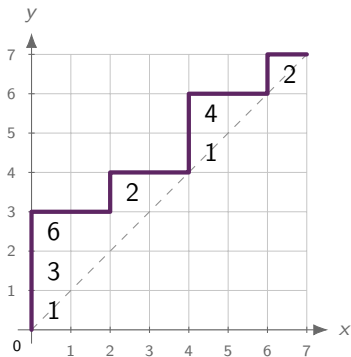
The north steps are numbered from bottom to top.

$$i < j, \quad a_i = a_j, \quad w_i < w_j.$$

Applying the same test to all row pairs gives

$$\boxed{\text{dinv}_{\text{primary}}(P) = 4}.$$

Counting one secondary dinv



The arrow records one contributing pair, not the full list of pairs.

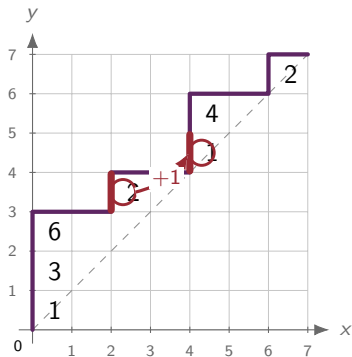
$$i < j, \quad a_i = a_j + 1, \quad w_i > w_j.$$

A secondary inversion connects two adjacent diagonals.

We test the pair

$$(i, j) = (4, 5).$$

Counting one secondary dinv



The arrow records one contributing pair, not the full list of pairs.

$$i < j, \quad a_i = a_j + 1, \quad w_i > w_j.$$

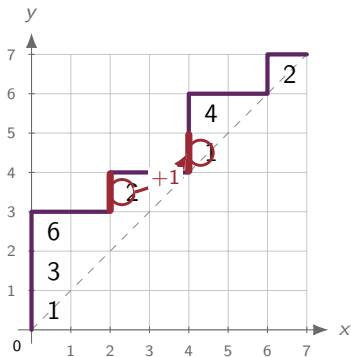
For rows 4 and 5,

$$a_4 = 1 = a_5 + 1,$$

$$w_4 = 2 > 1 = w_5.$$

Therefore (4, 5) contributes one dinv.

Counting one secondary dinv



The arrow records one contributing pair, not the full list of pairs.

$$i < j, \quad a_i = a_j + 1, \quad w_i > w_j.$$

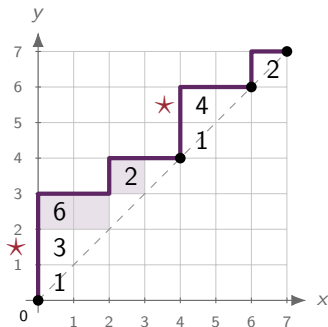
Applying the same test to all row pairs gives

$$\text{dinv}_{\text{secondary}}(P) = 6,$$

$$\boxed{\text{dinv}(P)} = 4 + 6 = 10.$$

Decorating two double rises

A double rise is an index i with $a_i = a_{i-1} + 1$.



$$r(\pi) = \{2, 3, 6\}, \quad dr = \{2, 6\}, \quad k = 2.$$

Starred rows do not contribute to area:

$$\text{area}(\pi, dr) = 5 - a_2 - a_6 = 3.$$

The return blocks have sizes $(4, 2, 1)$.
Removing the starred rows gives

$$\text{dcomp}(\pi, dr) = (3, 1, 1) \models 5.$$

Hence this labelled path contributes

$$q^3 t^{10} x_1^2 x_2^2 x_3 x_4 x_6.$$

The compositional Delta theorem

D'Adderio–Mellit

For $\alpha \models n - k$,

$$\Theta_{e_k} \nabla C_\alpha(X; t) = \sum_{\substack{P \in \text{LD}(n)^{*k} \\ \text{dcomp}(P) = \alpha}} q^{\text{area}(P)} t^{\text{dinv}(P)} x^P.$$

- k is the number of decorated rises.
- At $k = 0$, this is the compositional shuffle theorem.
- Our path has $n = 7$, $k = 2$, $\alpha = (3, 1, 1)$, and contributes $q^3 t^{10} x_1^2 x_2^2 x_3 x_4 x_6$.

The spaces $\mathcal{P}(\ell)$

For $\ell \geq 0$,

$$\mathcal{P}(\ell) := \mathbb{Q}(q, t)[x_1, \dots, x_\ell] \otimes \text{Sym}(x_{\ell+1}, x_{\ell+2}, \dots).$$

An element may be nonsymmetric in $x_1, \dots, x_\ell$ but is symmetric in the tail.

space	example	meaning
$\mathcal{P}(0)$	$s_{(2,1)}[X]$	fully symmetric
$\mathcal{P}(1)$	$x_1^2 h_1[x_2 + x_3 + \dots]$	x_1 distinguished
$\mathcal{P}(2)$	$x_1 x_2^2 e_2[x_3 + x_4 + \dots]$	x_1, x_2 distinguished

The elementary Demazure operator

Let s_i swap x_i and x_{i+1} . The Demazure operator is

$$\pi_i(f) := \frac{(1 - s_i)(x_i f)}{x_i - x_{i+1}} = \frac{x_i f - s_i(x_i f)}{x_i - x_{i+1}}.$$

Three local calculations are

$$\pi_i(1) = \frac{x_i - x_{i+1}}{x_i - x_{i+1}} = 1,$$

$$\pi_i(x_i) = \frac{x_i^2 - x_{i+1}^2}{x_i - x_{i+1}} = x_i + x_{i+1}, \quad \pi_i(x_{i+1}) = 0.$$

For a reduced word $w = s_{i_1} \cdots s_{i_d}$, set $\pi_w = \pi_{i_1} \cdots \pi_{i_d}$.

Stable Weyl symmetrization

Let $w_0^{(\ell, N)}$ be the longest permutation of $x_\ell, x_{\ell+1}, \dots, x_N$. For $F \in \mathcal{P}(\ell)$, define

$$\pi_\ell F[X] := \lim_{N \rightarrow \infty} \pi_{w_0^{(\ell, N)}} F[x_1, \dots, x_N, 0, 0, \dots].$$

Then

$$\pi_\ell : \mathcal{P}(\ell) \longrightarrow \mathcal{P}(\ell - 1),$$

and therefore

$$\mathcal{P}(\ell) \xrightarrow{\pi_\ell} \mathcal{P}(\ell - 1) \longrightarrow \dots \longrightarrow \mathcal{P}(0) = \text{Sym}[X].$$

The sequence converges strongly; π_ℓ is the stable Demazure symmetrizer.

Calculation: $\pi_1(x_1) = h_1[X]$

First use two variables:

$$\pi_1(x_1) = x_1 + x_2.$$

Add a third variable and symmetrize the new tail:

$$\pi_2\pi_1(x_1) = \pi_2(x_1 + x_2) = x_1 + x_2 + x_3.$$

Continuing gives

$$\pi_{w_0^{(1,N)}}(x_1) = x_1 + x_2 + \cdots + x_N = h_1[X_N].$$

Hence

$$\pi_1(x_1) = \lim_{N \rightarrow \infty} h_1[X_N] = h_1[X].$$

Calculation: $\pi_1\pi_2(x_1x_2) = e_2[X]$

At $N = 3$, first absorb x_2 into its tail:

$$\pi_2(x_1x_2) = x_1(x_2 + x_3).$$

Then

$$\begin{aligned}\pi_1(x_1(x_2 + x_3)) &= \frac{x_1^2(x_2 + x_3) - x_2^2(x_1 + x_3)}{x_1 - x_2} \\ &= x_1x_2 + x_1x_3 + x_2x_3 = e_2[X_3].\end{aligned}$$

The remaining longest-word factors fix this symmetric polynomial. Therefore

$$\pi_1\pi_2(x_1x_2) = \sum_{1 \leq i < j} x_ix_j = e_2[X].$$

Π_ℓ : nonsymmetric plethysm

$$\boxed{\Pi_\ell : \mathcal{P}(\ell) \longrightarrow \mathcal{P}(\ell)}$$

For $f \in \mathbb{Q}(q, t)[x_1, \dots, x_\ell]$ and symmetric g , define

$$\Pi_\ell(f(x_1, \dots, x_\ell)g[X]) := g\left[\frac{X}{1-t}\right] \text{pol}\left(\frac{f(x_1, \dots, x_\ell)}{\prod_{1 \leq i < j \leq \ell} (1 - tx_i/x_j)}\right).$$

- Π_ℓ is the ℓ -nonsymmetric plethysm map.
- It is an invertible $\mathbb{Q}(q, t)$ -linear map on $\mathcal{P}(\ell)$.
- pol is the polynomial-part projection in the Demazure-character expansion.
- It preserves ℓ ; unlike π_ℓ , it does not forget a flag.

Blasiak–Haiman–Morse–Pun–Seelinger. The operator is Π_ℓ , not the symmetric Π .

Integral and modified Macdonald bases

Put $\mu = (\eta; \lambda)_+$. The four bases fit into the commutative square

$$\begin{array}{ccc}
 \check{J}_{\eta|\lambda}(X; q, t) & \xrightarrow{\Pi_\ell} & \check{H}_{\eta|\lambda}(X; q, t) \\
 \text{integral-form stable nonsymmetric Macdonald} & & \text{modified nonsymmetric Macdonald} \\
 \downarrow \text{stable Hecke } \widehat{S}_1 \cdots \widehat{S}_\ell & & \downarrow \text{stable Weyl } \pi_1 \cdots \pi_\ell \\
 q^{n(\mu^*)} J_\mu(X; q^{-1}, t) & \xrightarrow{F[X] \mapsto F[X/(1-t)]} & \omega \widetilde{H}_\mu(X; q, t)
 \end{array}$$

Thus $\check{H}_{\eta|\lambda} = \Pi_\ell \check{J}_{\eta|\lambda}$, exactly as $\widetilde{H}_\mu$ is obtained from the integral form by symmetric plethysm.

The left vertical map is Hecke symmetrization; after Π_ℓ it becomes Weyl symmetrization on the modified side.

Ordinary LLT polynomials: the combinatorial definition

Let $\nu = (\nu^{(1)}, \dots, \nu^{(r)})$ be a tuple of skew diagrams. For cells $u \in \nu^{(i)}$ and $v \in \nu^{(j)}$, with content $c(\cdot)$, say that u attacks v when

$$i < j \text{ and } c(v) = c(u), \quad \text{or} \quad i > j \text{ and } c(v) = c(u) + 1.$$

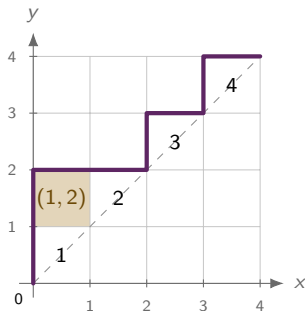
For a semistandard filling T , let $\text{inv}(T)$ count attacking pairs with $T(u) > T(v)$. Then

$$\text{LLT}_\nu(X; t) := \sum_{T \in \text{SSYT}(\nu)} t^{\text{inv}(T)} x^T.$$

This generating function is symmetric in X .

Lascoux–Leclerc–Thibon; Grojnowski–Haiman; Blasiak–Haiman–Morse–Pun–Seelinger.

An ordinary LLT example with $n = 4$



$$\rho = \text{NNEENENE}, \quad a(\rho) = (0, 1, 0, 0).$$

The shaded box gives the single attacking pair

$$\text{Area}(\rho) = \{(1, 2)\}, \quad \text{inv}_\rho(w) = [w_1 > w_2].$$

Therefore

$$\begin{aligned} \text{LLT}_\rho(X; t) &= \left(\sum_{a, b \geq 1} t^{[a > b]} x_a x_b \right) h_1[X]^2 \\ &= (h_2[X] + t e_2[X]) h_1[X]^2. \end{aligned}$$

Flagged row LLTs: unsigned and signed

Let $\hat{\pi} \in D_\ell(n)$ be an ℓ -partial Dyck path with marking Σ . Every flagged word satisfies

$$T(r) \leq r \quad (1 \leq r \leq \ell).$$

Unsigned row words. For $T : [n] \rightarrow \mathbb{Z}_{>0}$,

$$T(j) \leq T(i) \quad ((i, j) \in \Sigma),$$

$$\text{inv}(T) = \#\{(i, j) \in \text{Area}(\hat{\pi}) : T(j) \leq T(i)\},$$

$$G_\ell(\hat{\pi}, \Sigma) = \sum_{T \in \text{FSW}(\hat{\pi}, \Sigma)} t^{\text{inv}(T)} x^T.$$

Signed row super words. Use

$$\mathcal{A} = \{1 < \bar{1} < 2 < \bar{2} < \dots\}.$$

A marked or attacking pair contributes when

$$T(j) < T(i), \quad \text{or } T(i) = T(j) \text{ is positive.}$$

If $m(T)$ counts barred letters, then

$$G_\ell^\pm(\hat{\pi}, \Sigma) = \sum_{T \in \text{FSW}^\pm(\hat{\pi}, \Sigma)} t^{\text{inv}(T)} (-t)^{m(T)} x^{|T|}.$$

From flagged row LLTs to ordinary LLT polynomials

Theorem (Blasiak–Haiman–Morse–Pun–Seelinger)

For $\hat{\pi} \in D_\ell(n)$ with marking Σ ,

$$\Pi_\ell G_\ell^\pm(\hat{\pi}, \Sigma) = G_\ell(\hat{\pi}, \Sigma).$$

If $\pi = N^\ell \hat{\pi}$, then

$$\pi_1 \cdots \pi_\ell G_\ell(\hat{\pi}, \Sigma) = \omega \text{ LLT}_{\pi, \Sigma}(X; t).$$

Thus the two operations have different jobs:

signed $\xrightarrow{\Pi_\ell}$ unsigned flagged $\xrightarrow{\pi_1 \cdots \pi_\ell}$ ordinary LLT.

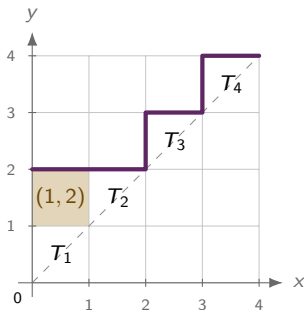
The factor ω comes from the row convention.

Qiu–Zhang, Theorem 2.5.7 and Proposition 2.5.8.

A size-4 example in $\mathcal{P}(2)$

Restore the path $\rho = N^2\hat{\pi}$ with

$$\hat{\pi} = \text{EENENE} \in D_2(4), \quad \Sigma = \emptyset, \quad \text{Area}(\hat{\pi}) = \{(1, 2)\}.$$



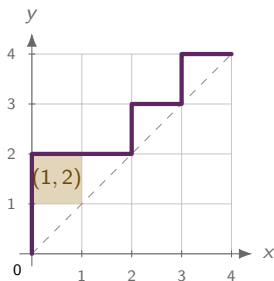
The flags give $T_1 = 1$ and $T_2 \in \{1, 2\}$. Since $\text{inv}(T) = [T_2 \leq T_1]$,

T_2	1	2
weight	tx_1^2	x_1x_2

The two free positions contribute $h_1[X]^2$. Hence

$$G_2(\hat{\pi}, \emptyset) = (tx_1^2 + x_1x_2)h_1[X]^2.$$

Weyl symmetrization restores the ordinary LLT



$$\rho = \text{NNEENENE}.$$

From the ordinary calculation,

$$\text{LLT}_\rho = (h_2 + te_2)h_1^2.$$

Start from the flagged answer:

$$G_2 = (tx_1^2 + x_1x_2)h_1^2.$$

Stable Weyl symmetrization gives

$$\pi_2(x_1x_2) = x_1(h_1[X] - x_1),$$

$$\pi_1\pi_2(x_1x_2) = h_1[X]^2 - h_2[X] = e_2[X],$$

$$\pi_1\pi_2(x_1^2) = h_2[X].$$

Since $h_1[X]^2$ is already symmetric,

$$\begin{aligned} \pi_1\pi_2 G_2 &= (th_2 + e_2)h_1^2 \\ &= \omega((h_2 + te_2)h_1^2) \\ &= \boxed{\omega \text{ LLT}_\rho(X; t)}. \end{aligned}$$

Where does $\mathcal{M}^Q$ come from?

The Dyck path algebra acts levelwise on

$$V_\ell = \mathbb{Q}(q, t)[y_1, \dots, y_\ell] \otimes \text{Sym}[X], \quad V_* = \bigoplus_{\ell \geq 0} V_\ell.$$

Carlsson–Mellit constructed an involution $\mathcal{N}$ on V_* with

$$\mathcal{N}|_{V_0} = \nabla' \overline{\omega}.$$

D'Adderio–Mellit extended $\Theta(u)\nabla'\overline{\omega}$ to every level by

$$\boxed{\mathcal{M} := \tau_{u, \ell}^* \mathcal{N} (\tau_{qtu, \ell}^*)^{-1}}.$$

In our conventions,

$$\mathcal{M}^Q = Q_\ell^{-1} \mathcal{M} Q_\ell, \quad \mathcal{M}_k^Q = (-1)^k \langle u^k \rangle \mathcal{M}^Q,$$

and, since $Q_0 = \text{id}$,

$$\boxed{\mathcal{M}^Q|_{V_0} = \Theta(u)\nabla'\overline{\omega}, \quad \mathcal{M}_k^Q|_{V_0} = \Theta_{e_k}\nabla'\overline{\omega}}.$$

ℓ is the module level; k indexes a coefficient of u . Here $\mathcal{M}$ is an operator, not the scalar $M = (1 - q)(1 - t)$.

The algebraic object $M_{\alpha,k}^Q$

Let $\alpha = (\alpha_1, \dots, \alpha_\ell) \models n - k$. In the paper,

$$M_{\alpha,k}^Q := (-1)^n P_\ell^{-1} t^{\ell-|\alpha|} \mathcal{M}_k^Q \left(y_1^{\alpha_1-1} \cdots y_\ell^{\alpha_\ell-1} d_+^\ell(1) \right),$$

where

$$\mathcal{M}^Q = Q_\ell^{-1} \mathcal{M} Q_\ell, \quad \mathcal{M}_k^Q = (-1)^k \langle u^k \rangle \mathcal{M}^Q.$$

Theorem (Qiu–Zhang)

For $\alpha \models n - k$ of length ℓ ,

$$M_{\alpha,k}^Q = \sum_{(\pi, dr) \in D(\alpha)^{*k}} q^{\text{area}(\pi, dr)} G_{\ell}^{\pm}(\hat{\pi}', \Sigma_{\pi}).$$

The right-hand side is a signed flagged row LLT expansion indexed by decorated Dyck paths of diagonal composition α .

Qiu–Zhang; D’Adderio–Mellit; Blasiak–Haiman–Morse–Pun–Seelinger.

The nonsymmetric compositional Delta theorem

Theorem (Qiu–Zhang)

For $\alpha \models n - k$ with $\ell(\alpha) = \ell$,

$$\Pi_\ell M_{\alpha,k}^Q = \sum_{(\pi, dr) \in D(\alpha)^{*k}} \sum_{w \in \text{FWPF}'_\alpha(\pi)} q^{\text{area}(\pi, dr)} t^{\text{dinv}'(\pi, w)} x^{\text{content}(w)}.$$

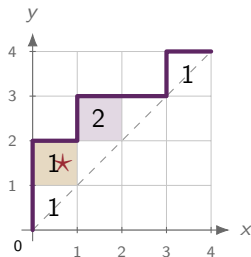
Here w is weakly decreasing upward in every vertical run, and the label on the j -th diagonal north step is at most j . The statistic is

$$\text{dinv}'(\pi, w) = \#\{(i, j) : i \text{ attacks } j, w_i \geq w_j\}.$$

Thus the right-hand side is a direct area–dinv' sum, with no flagged-LLT abbreviation.

Qiu–Zhang, Corollary 3.2.3.

Example: one term in the weak formula



$$\pi = \text{NNENEENE}, \quad dr = \{2\},$$

$$a(\pi) = (0, 1, 1, 0), \quad \text{dcomp}(\pi, dr) = (2, 1).$$

The decorated second row is omitted, so

$$\text{area}(\pi, dr) = (0 + 1 + 1 + 0) - 1 = 1.$$

Take $w = (1, 1, 2, 1)$. The weak run condition is $w_1 = 1 \geq 1 = w_2$, and the diagonal flags are $w_1 \leq 1$, $w_4 \leq 2$. All attacks can now be checked:

(i, j)	area relation	$w_i \geq w_j$
$(1, 4)$	$0 = 0$	✓
$(2, 3)$	$1 = 1$	×
$(2, 4)$	$1 = 0 + 1$	✓
$(3, 4)$	$1 = 0 + 1$	✓

Thus $\text{dinv}'(\pi, w) = 3$, and this labelled path contributes

$$qt^3 x_1^3 x_2.$$

Operators for the column and modified formulations

Let $\mu = (\eta; \lambda)_+$. The two nonsymmetric nabla operators are

$$\begin{aligned}\bar{\nabla} \check{J}_{\eta|\lambda} &= (-1)^{|\mu|} T_{\mu} \check{J}_{\eta|\lambda}, \\ \nabla \check{H}_{\eta|\lambda} &= (-1)^{|\mu|} T_{\mu} \check{H}_{\eta|\lambda}.\end{aligned}$$

Our two nonsymmetric τ^* operators are

$$\begin{aligned}\overline{\tau_{u,\ell}^*} &:= P_{\ell}^{-1} \tau_{u,\ell}^* P_{\ell} = \sum_{r \geq 0} (-u)^r e_r \left[\frac{X}{q-1} \right], \\ \underline{\tau_{u,\ell}^*} &:= \Pi_{\ell} \overline{\tau_{u,\ell}^*} \Pi_{\ell}^{-1}.\end{aligned}$$

Finally,

$$\vartheta F[X; q, t] = F[X; q^{-1}, t^{-1}], \quad [A(u)]_k := (-1)^k [u^k] A(u).$$

The signed column theorem

Theorem (Qiu–Zhang)

For $\alpha \models n - k$ with $\ell(\alpha) = \ell$,

$$\begin{aligned} & (-1)^{n+k} (-t)^{\ell - |\alpha|} \vartheta \left(\left[\left(\overline{\tau_{qtu,\ell}^*} \right)^{-1} \bar{\nabla}^{-1} \left(\overline{\tau_{qtu,\ell}^*} \right)^{-1} \right]_k x_1^{\alpha_1} \cdots x_\ell^{\alpha_\ell} \right) \\ &= \sum_{(\pi, dr) \in D(\alpha)^{*k}} \sum_{T \in \widetilde{\text{FSW}}^\pm(\hat{\pi}', \Sigma_\pi)} q^{\text{area}(\pi, dr)} t^{\widetilde{\text{inv}}(T)} (-t)^{-m(T)} x^{|T|}. \end{aligned}$$

Here T is a signed column super word. For a marked or attacking pair, the column condition is

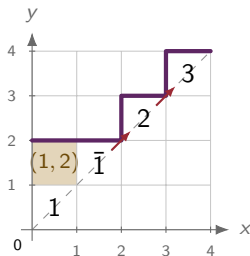
$$T(j) > T(i), \quad \text{or } T(i) = T(j) \text{ is a barred letter.}$$

The same predicate defines $\widetilde{\text{inv}}(T)$, while $m(T)$ counts barred letters.

Qiu–Zhang, Corollary 4.2.5 and Definition 2.5.4.

Example: one term in the signed column formula

$$\sum_{(\pi, dr) \in D(\alpha)^{*k}} \sum_{T \in \widetilde{\text{FSW}}^{\pm}(\hat{\pi}', \Sigma_{\pi})} q^{\text{area}(\pi, dr)} t^{\widetilde{\text{inv}}(T)} (-t)^{-m(T)} x^{|T|}.$$



Use the same $n = 4$, $k = 1$ decorated path, so $\text{area}(\pi, dr) = 2$, and take $T = (1, \bar{1}, 2, 3)$. The flags hold; the marked conditions are $2 > \bar{1}$ and $3 > 2$. The unique area pair contributes because $\bar{1} > 1$:

$$\widetilde{\text{inv}}(T) = 1, \quad m(T) = 1, \quad x^{|T|} = x_1^2 x_2 x_3.$$

Therefore this one super word contributes

$$q^2 t (-t)^{-1} x_1^2 x_2 x_3 = -q^2 x_1^2 x_2 x_3.$$

$$\begin{aligned} \hat{\pi}' &= \text{EENENE}, & \text{Area}(\hat{\pi}') &= \{(1, 2)\}, \\ \Sigma_{\pi} &= \{(2, 3), (3, 4)\}. \end{aligned}$$

The final modified theorem

Define

$$C_\alpha := (-t)^{|\alpha|-\ell} \prod_\ell (x_1^{\alpha_1} \cdots x_\ell^{\alpha_\ell}).$$

Theorem (Qiu-Zhang)

For $\alpha \models n - k$ with $\ell(\alpha) = \ell$,

$$\begin{aligned} & (-1)^{n+k} \vartheta \left(\left[\left(\tau_{qtu, \ell}^* \right)^{-1} \nabla^{-1} \left(\tau_{qtu, \ell}^* \right)^{-1} \right]_k C_\alpha \right) \\ &= \sum_{(\pi, dr) \in D(\alpha)^{*k}} \sum_{w \in \text{FWPF}_\alpha(\pi)} q^{\text{area}(\pi, dr)} t^{\text{dinv}(\pi, w)} x^{\text{content}(w)}. \end{aligned}$$

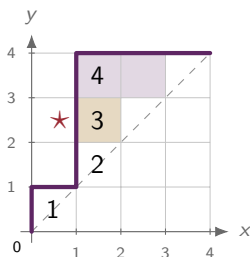
Here labels are strictly increasing upward in every vertical run, with the same diagonal flags, and

$$\text{dinv}(\pi, w) = \#\{(i, j) : i \text{ attacks } j, w_i < w_j\}.$$

Qiu-Zhang, Corollary 4.2.7.

Example: one term in the final modified formula

$$\sum_{(\pi, dr) \in D(\alpha)^{*k}} \sum_{w \in \text{FWPF}_{\alpha}(\pi)} q^{\text{area}(\pi, dr)} t^{\text{dinv}(\pi, w)} x^{\text{content}(w)}.$$



Take $w = (1, 2, 3, 4)$. On the long vertical run, $2 < 3 < 4$, and the diagonal flags are

$$w_1 = 1 \leq 1, \quad w_2 = 2 \leq 2.$$

The only attack is $(1, 2)$, and

$$w_1 = 1 < 2 = w_2, \quad \text{dinv}(\pi, w) = 1.$$

Therefore this one labelled decorated path contributes

$$q^2 t x_1 x_2 x_3 x_4.$$

$$\begin{aligned} \pi &= \text{NENNEEE}, & dr &= \{3\}, \\ \text{dcomp}(\pi, dr) &= (1, 2), & \text{area}(\pi, dr) &= 2. \end{aligned}$$

Recovering the symmetric compositional Delta theorem

Apply $\pi_1 \cdots \pi_\ell$ to the final modified identity.

Operator side

$$\begin{aligned} \pi_1 \cdots \pi_\ell (\text{left-hand side}) \\ = \Theta_{e_k} \nabla C_\alpha(X; t). \end{aligned}$$

The modified nonsymmetric operators descend to their symmetric counterparts.

Combinatorial side

$$\begin{aligned} \pi_1 \cdots \pi_\ell \left(\sum_{(\pi, dr)} \sum_{w \in \text{FWPF}_\alpha(\pi)} q^{\text{area}} t^{\text{dinv}_X \text{content}(w)} \right) \\ = \sum_{\substack{P \in \text{LD}(n)^{*k} \\ \text{dcomp}(P) = \alpha}} q^{\text{area}(P)} t^{\text{dinv}(P)}_X P. \end{aligned}$$

The flags disappear and the first ℓ north steps are restored.

The two images give exactly the symmetric compositional Delta theorem.

D'Adderio–Mellit; Qiu–Zhang, Proposition 4.3.1.

Main results and consequences

- ➊ **Refinement of D'Adderio–Mellit.** Weyl symmetrization gives the compositional Delta theorem.
- ➋ **Extension of the nonsymmetric shuffle theorem.** The case $k = 0$ is the Blasiak–Haiman–Morse–Pun–Seelinger theory; the new formulas cover $k > 0$.
- ➌ **Four compatible formulations.** Signed row, unsigned row, signed column, and final modified forms describe the same Delta object.
- ➍ **Combinatorial formulas.** The unsigned forms use weak or strict flagged word parking functions.
- ➎ **Further questions.** The formulas suggest sharper positivity and structural phenomena in $\mathcal{P}(\ell)$.

The nonsymmetric identities retain information that is lost after symmetrization.

Proof outline

- 1 Start from the D'Adderio–Mellit decorated-path identity in the Dyck path algebra.
- 2 Translate conventions and transport the identity to $\mathcal{P}(\ell)$ through P_ℓ^{-1} .
- 3 Recognize each path term as $G_\ell^\pm(\hat{\pi}', \Sigma_\pi)$.
- 4 Apply Π_ℓ to obtain the unsigned row and weak flagged-word formulas.
- 5 Use the column transformation, nonsymmetric nabla, and the two τ^* operators for the final modified form.
- 6 Apply $\pi_1 \cdots \pi_\ell$ to recover the symmetric theorem.

The technical work is to verify the required intertwining relations.

Conjecture 5.1.1: stable-atom positivity

Stable atoms $\mathcal{A}_{\eta|\lambda}$ form a basis of $\mathcal{P}(\ell)$ whose full Weyl images are Schur functions or zero.

Conjecture 5.1.1

① Flagged row and column LLT polynomials are stable-atom positive.

② $\Pi_{\ell} M_{\alpha,k}^Q$ is stable-atom positive.

③ The element

$$(-1)^{n+k} \left[\underline{\tau_{qtu,\ell}^*}^{-1} \nabla^{-1} \underline{\tau_{qtu,\ell}^*}^{-1} \right]_k C_{\alpha}$$

is stable-atom positive.

Item (1), originally conjectured by BHMPSe, implies items (2) and (3).

Open Problem 5.2.1: nonsymmetric Theta calculus

In the symmetric setting, the Delta deformation is packaged by

$$\Theta_{e_k} \nabla' = \sum_{i=0}^k e_i^* \nabla' e_{k-i}^*.$$

Question

Is there an analogous structural identity in $\mathcal{P}(\ell)$ involving

$$(\bar{\nabla}, \bar{\Theta}_f) \quad \text{or} \quad (\nabla, \Theta_f)?$$

Such an identity would give a compact reformulation of the two main nonsymmetric Delta formulas.

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Thank you!